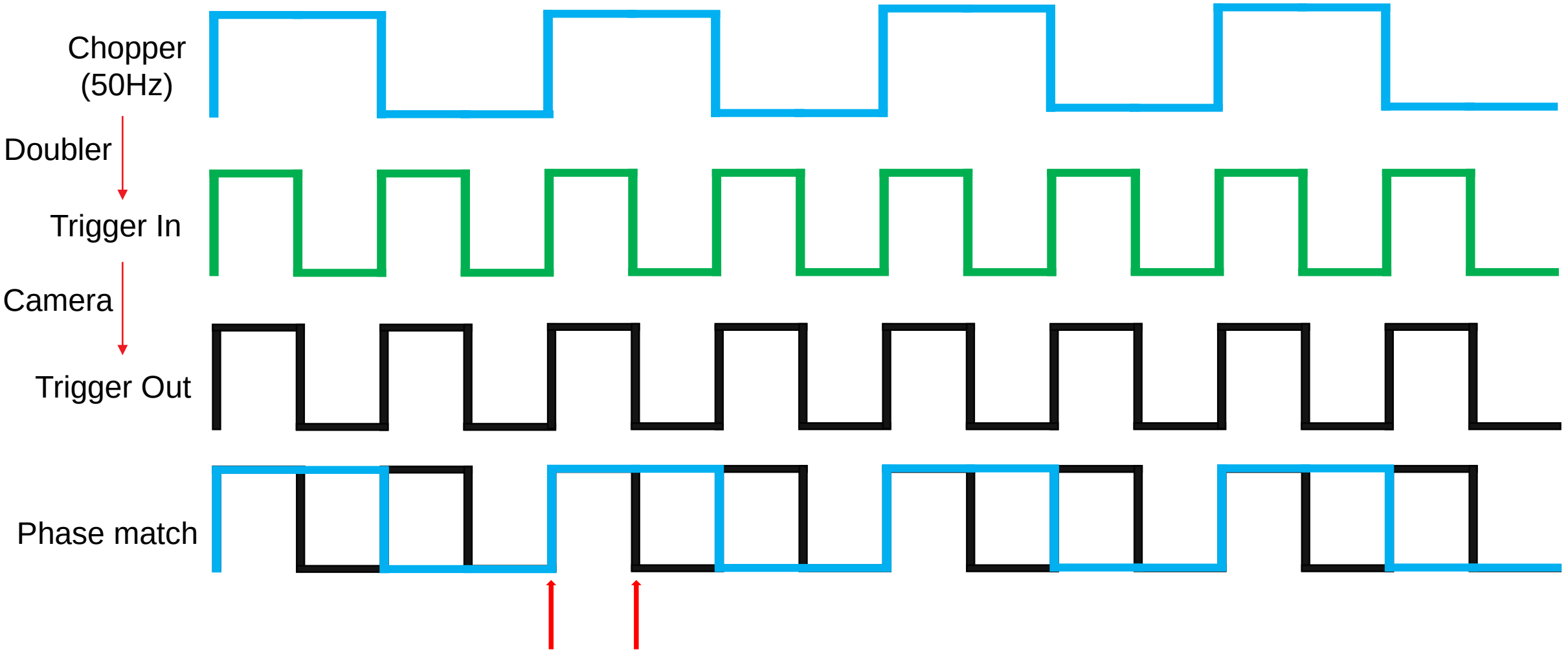
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Shutter Controlled Pump-probe

Speaker: Zhongqi Xiu

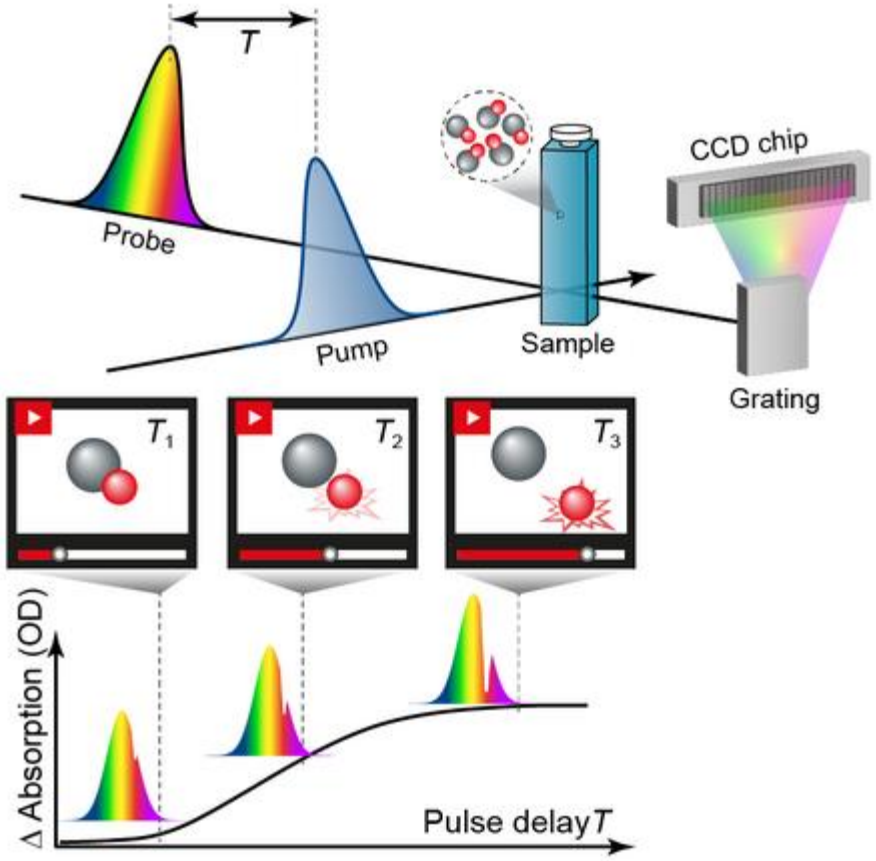
Timing Diagram of Chopper Pump-Probe

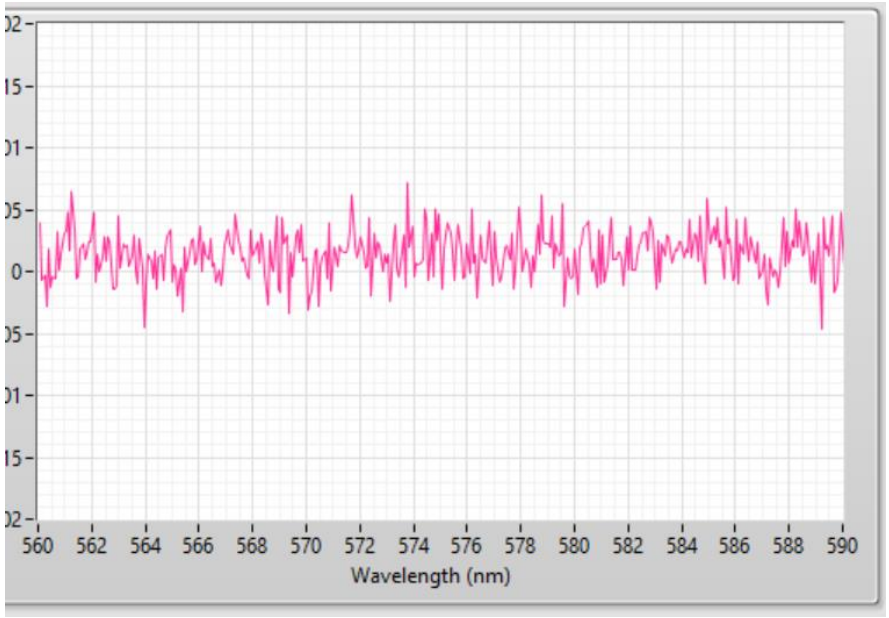
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Why change chopper to shutter

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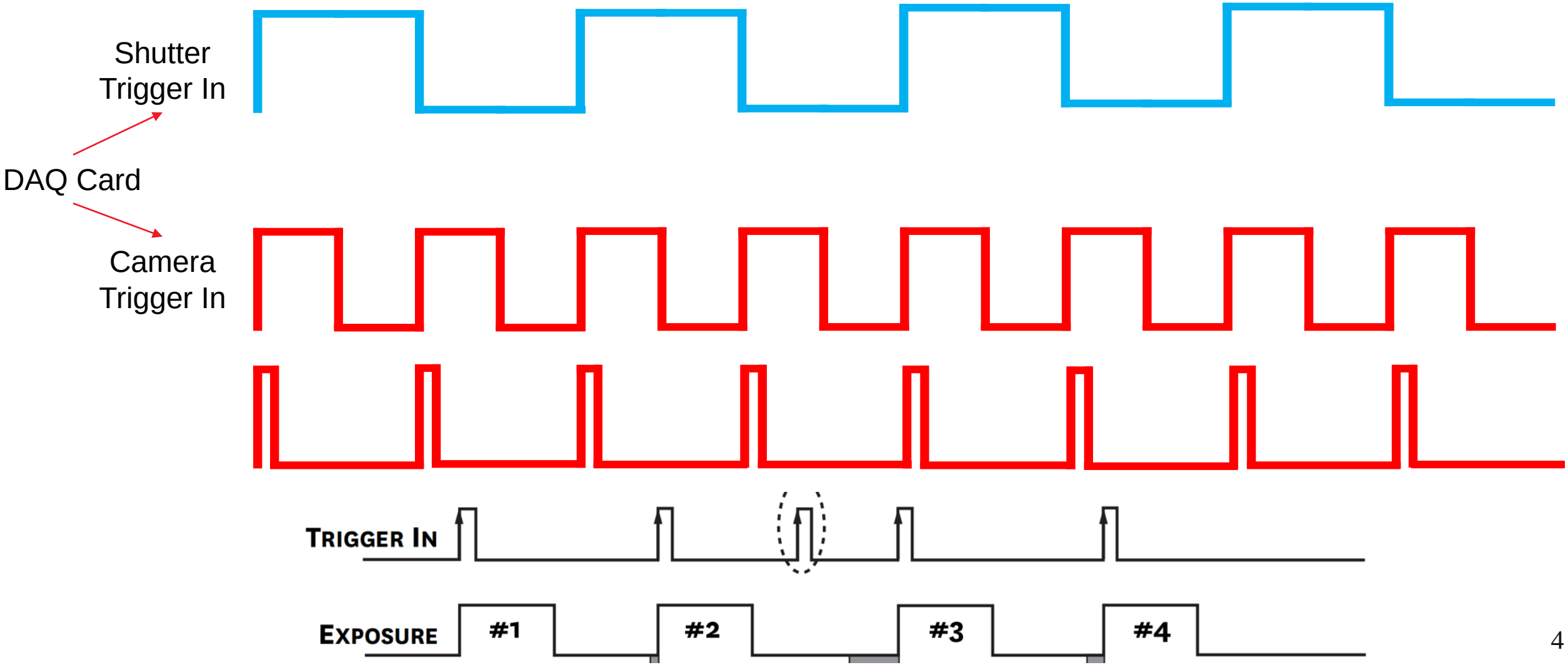


Increase integration time+ more frames --- higher SNR

Random noise (time and wavelength independent)

Timing Diagram of Shutter Pump-Probe

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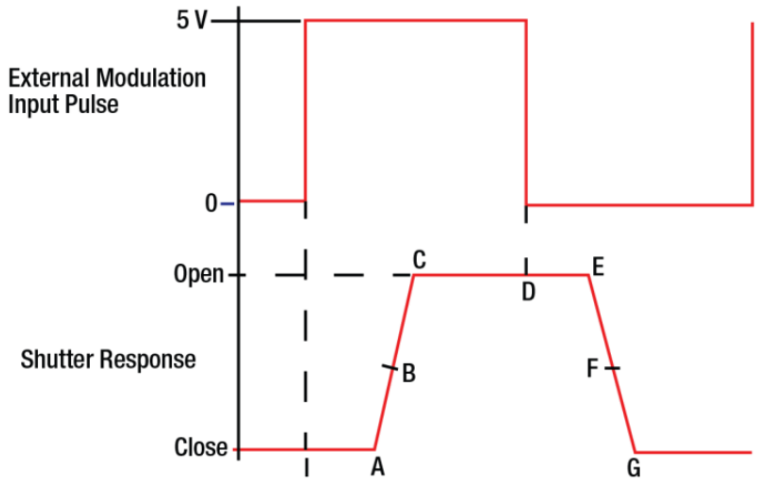


Timing Diagram of Shutter Pump-Probe

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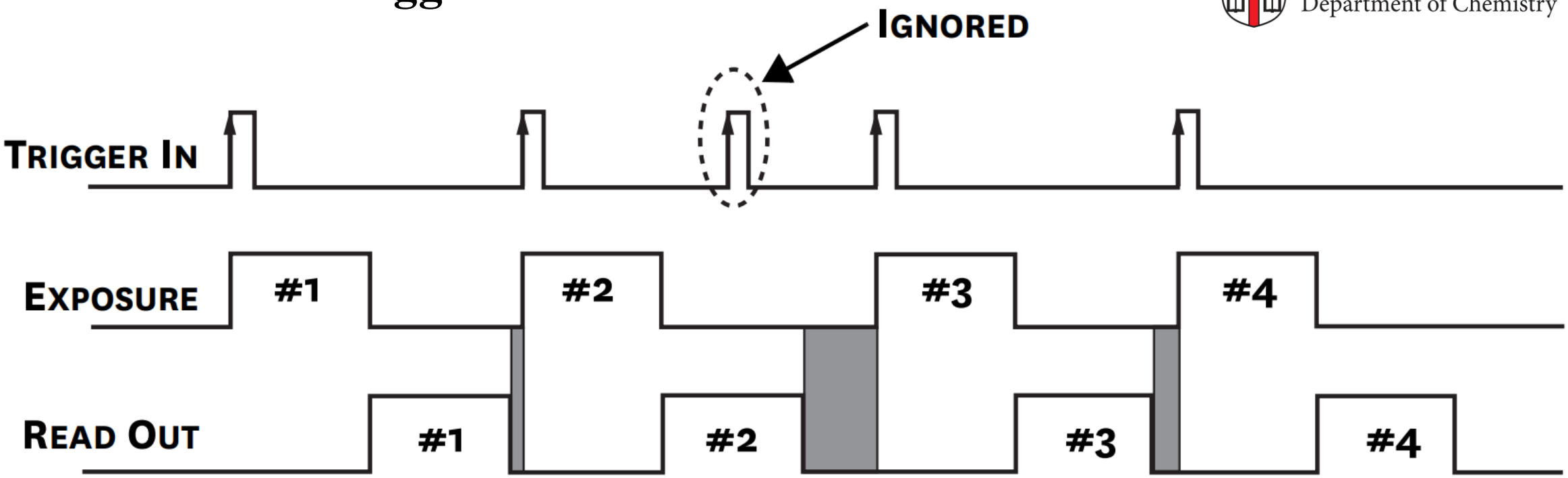


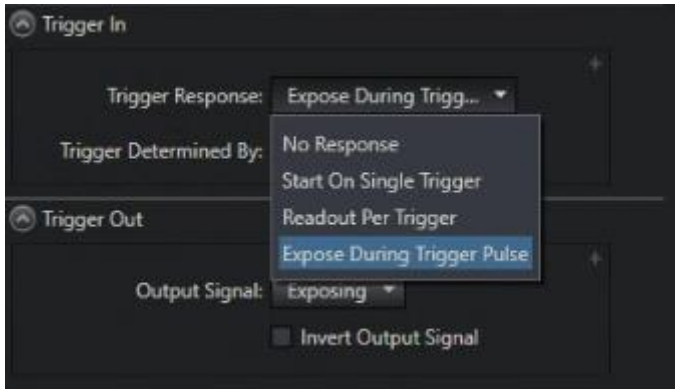


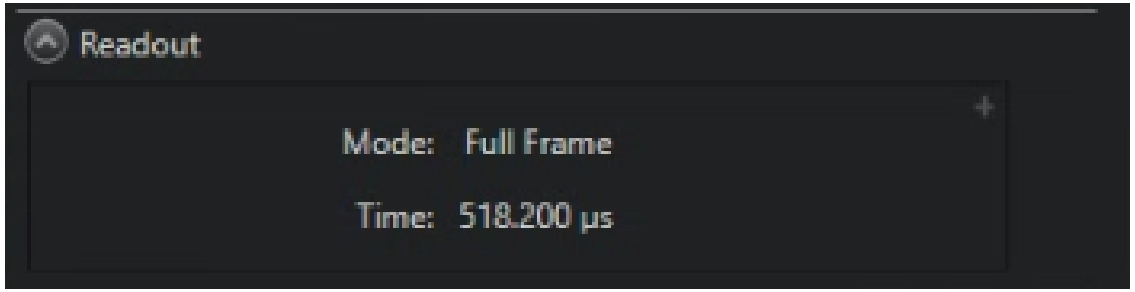
	Time (ms, Typ)	Time (ms, Max)	Definitions
I → A	11	13	Delay from input pulse rising edge to initiation of shutter opening.
A → C	10	12	Rising edge to 100% open.
I → C	21	25	Delay from input pulse rising edge to shutter fully opening.
D → E	15	17	Delay from fall of input pulse to initiation of shutter fully closing.
E → G	9	11	Falling edge to 100% closed.
D → G	24	26	Delay from fall of input drive to shutter fully closed.

“Readout Per Trigger” Mode

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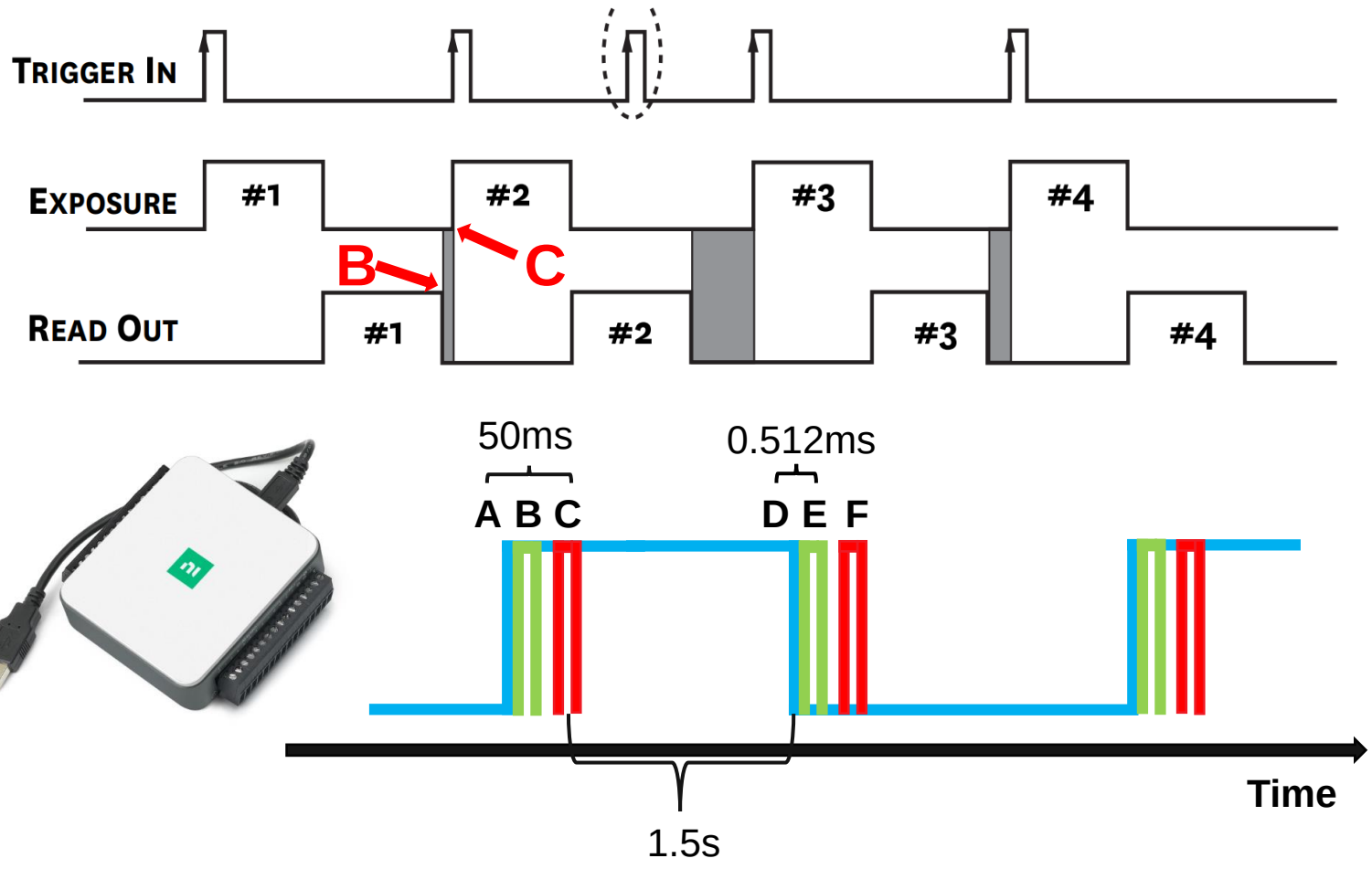






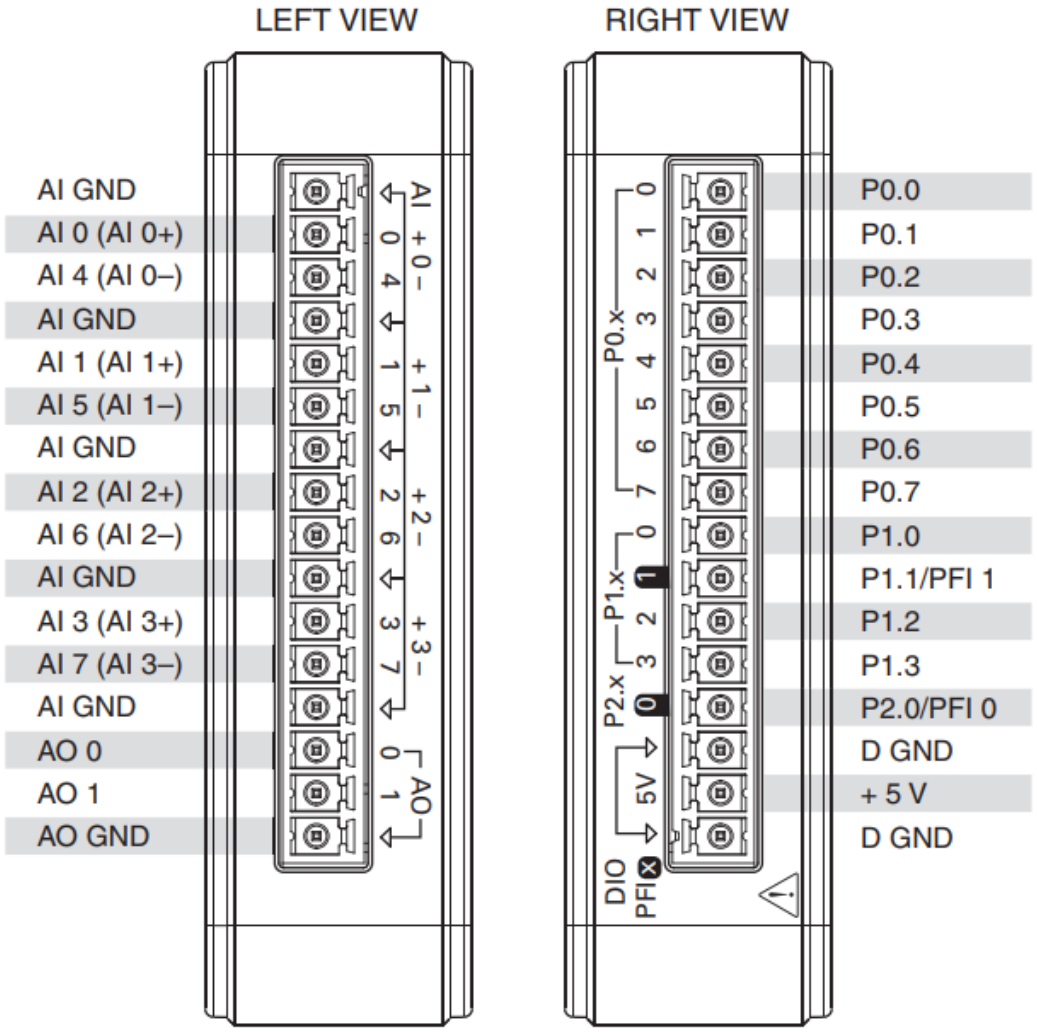
Timing Diagram On & Off

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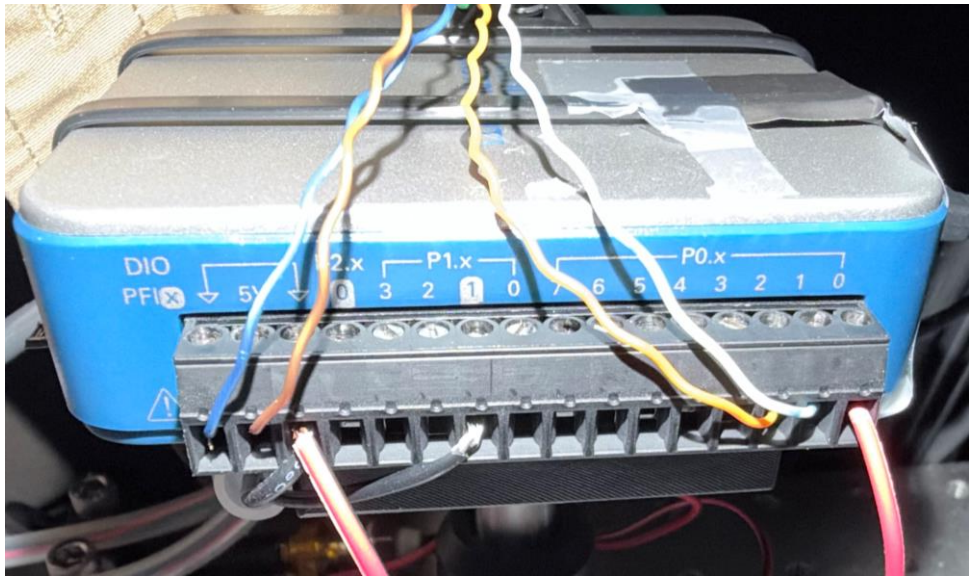


Time	Shutter	Camera
A	Receive TTL	Exposure End
B	Process	Read Out
C	Fully open	Exposure
D	Receive TTL	Exposure End

DAQ Card TTL Output

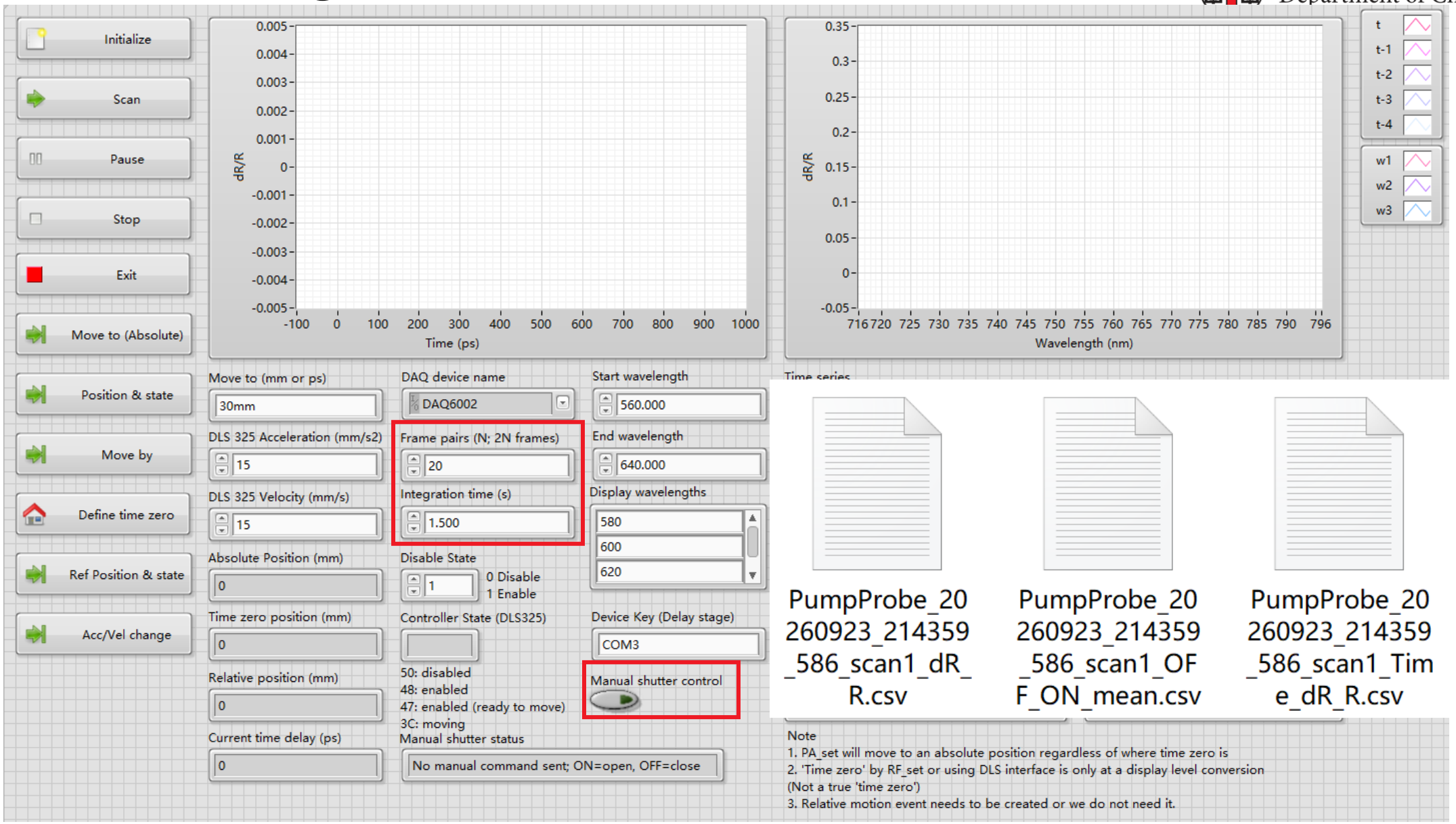


P2.0	D GND	Input or Output	**Port 2 Digital I/O Channel 0**—You can configure each signal individually as an input or output. Refer to the *Digital I/O* section for more information.

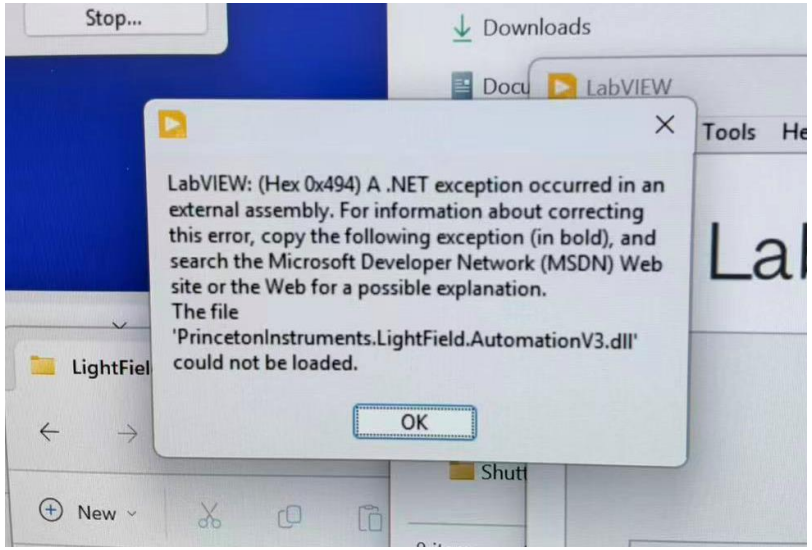


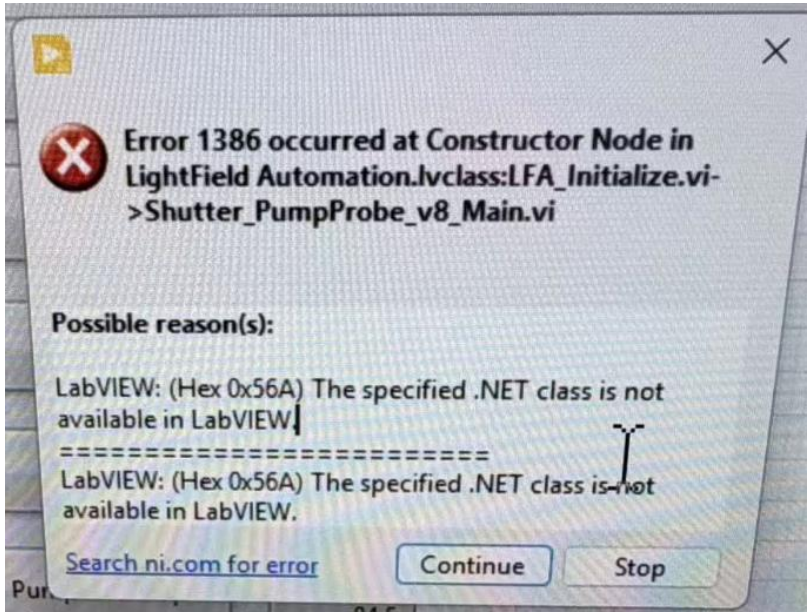
LabVIEW Program

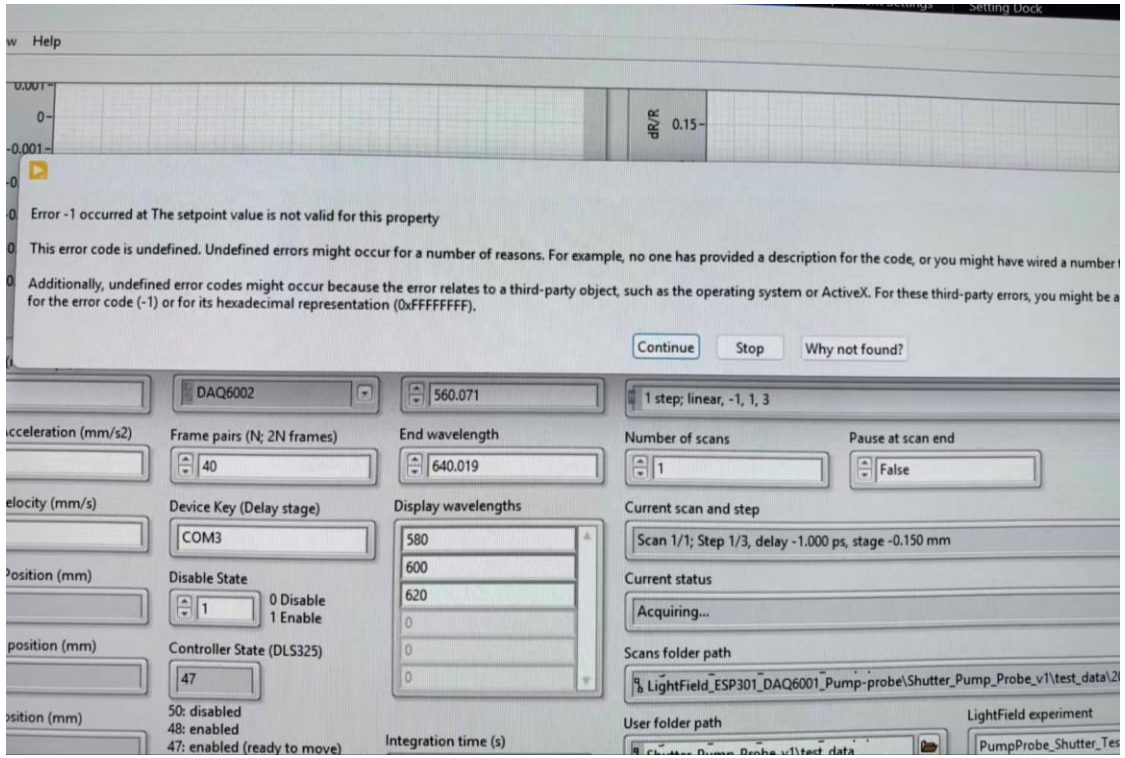
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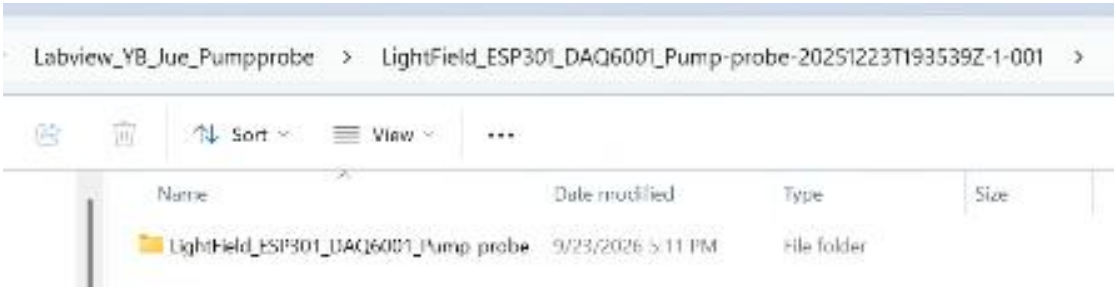


Bugs


Stop...
Downloads
LabVIEW
Tools
LightFiel
New
LabVIEW: (Hex 0x494) A .NET exception occurred in an external assembly. For information about correcting this error, copy the following exception (in bold), and search the Microsoft Developer Network (MSDN) Web site or the Web for a possible explanation.
The file 'PrincetonInstruments.LightField.AutomationV3.dll' could not be loaded.
OK

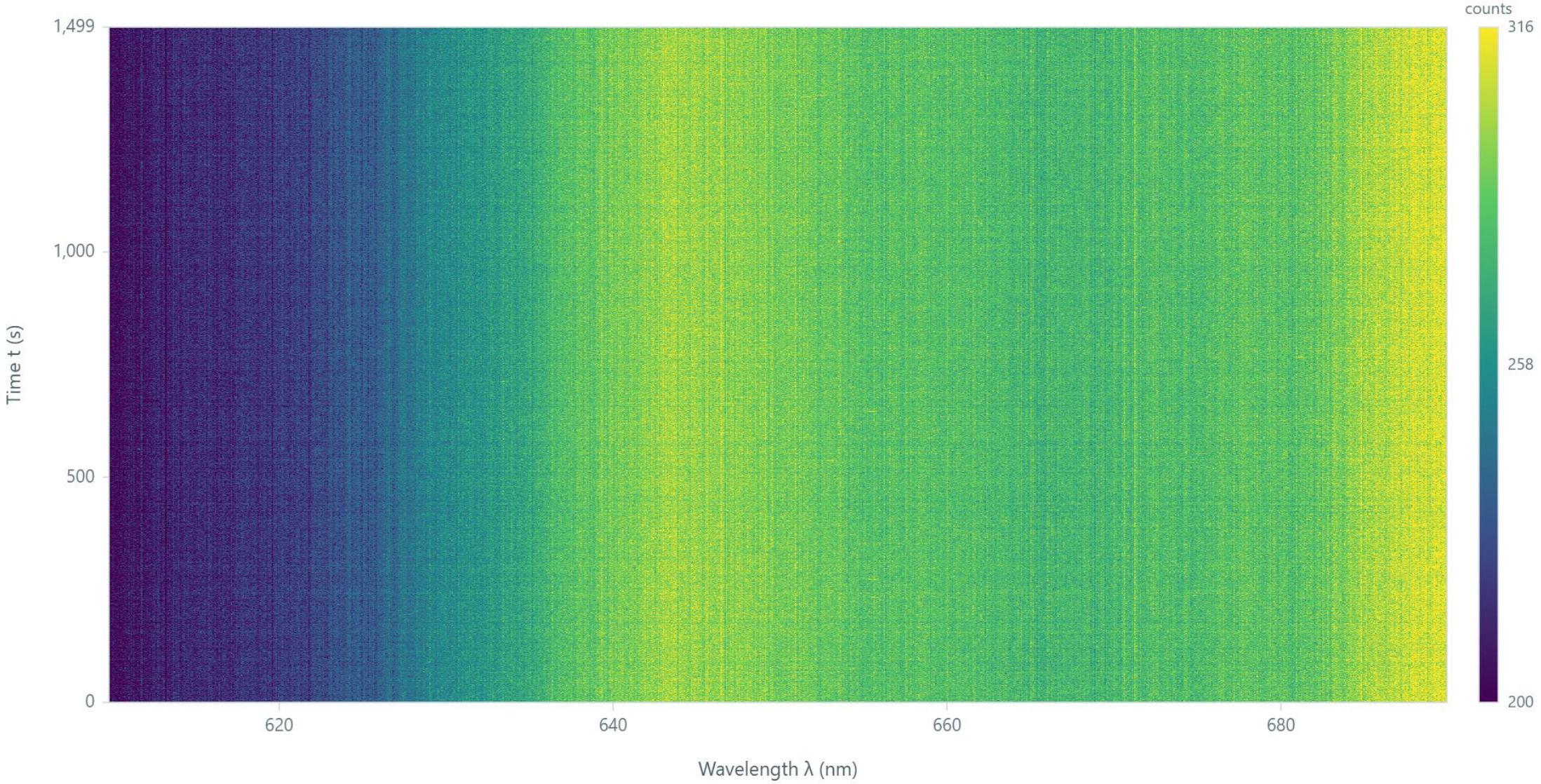

Error 1386 occurred at Constructor Node in LightField Automation.lvclass:LFA_Initialize.vi->Shutter_PumpProbe_v8_Main.vi
Possible reason(s):
LabVIEW: (Hex 0x56A) The specified .NET class is not available in LabVIEW.
==========
LabVIEW: (Hex 0x56A) The specified .NET class is not available in LabVIEW.
Search ni.com for error
Continue
Stop


Help
Error -1 occurred at The setpoint value is not valid for this property
This error code is undefined. Undefined errors might occur for a number of reasons. For example, no one has provided a description for the code, or you might have wired a number
Additionally, undefined error codes might occur because the error relates to a third-party object, such as the operating system or ActiveX. For these third-party errors, you might be a
for the error code (-1) or for its hexadecimal representation (0xFFFFFFFF).
Continue
Stop
Why not found?
DAQ6002
560.071
1 step; linear, -1, 1, 3
Frame pairs (N; 2N frames)
End wavelength
Number of scans
Pause at scan end
40
640.019
1
False
Device Key (Delay stage)
Display wavelengths
Current scan and step
COM3
580
600
620
Scan 1/1; Step 1/3, delay -1.000 ps, stage -0.150 mm
Disable State
Current status
1
0 Disable
1 Enable
Acquiring...
Controller State (DLS325)
Scans folder path
47
50: disabled
48: enabled
47: enabled (ready to move)
Integration time (s)
User folder path
LightField experiment


Labview_YB_Jue_Pumpprobe
LightField_ESP301_DAQ6001_Pump-probe-20251223T193539Z-1-001
Sort
View
Name
Date modified
Type
Size
LightField_ESP301_DAQ6001_Pump probe
9/23/2026 5:11 PM
File folder

White Light Fluctuation

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White Light Fluctuation

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